SOC HOME LAB

Security Operations Center Simulation

Attack Simulation & Detection Report

Phase 1 (No WAF) to Phase 2 (ModSecurity + OWASP CRS)

Date	April 26-27, 2026
Classification	Confidential - Lab Use Only
Network	192.168.75.0/24 | VMware Workstation
Target Application	OWASP Juice Shop v19.2.1
SIEM Platform	Wazuh v4.14.5
WAF	ModSecurity v3.0.12 + OWASP CRS 3.3.5
Attacker Platform	Kali Linux 2024 (192.168.75.128)
Document Version	v1.0 Final

Contents

1. Executive Summary

This report documents the design, deployment, and operational results of a Security Operations Center (SOC) home lab built on native software inside VMware Workstation VMs. The lab simulates a realistic enterprise threat environment covering active exploitation, detection, alerting, and response using industry-standard open-source tooling.

The exercise ran in two phases. Phase 1 had Kali Linux attacking OWASP Juice Shop with no application-layer defenses. Phase 2 deployed ModSecurity + OWASP CRS as a WAF and repeated the identical attacks. A Wazuh agent forwarded all logs to a centralized SIEM where custom MITRE ATT&CK-mapped rules generated real-time alerts.

1.1 Key Results Summary

Attack	Phase 1 - No WAF	Phase 2 - WAF Active
XSS Reflected	HTTP 200 - PASSED (not blocked)	HTTP 403 - BLOCKED
SQL Injection Auth Bypass	HTTP 200 - Admin JWT obtained	HTTP 403 - BLOCKED
Hydra Brute Force (102 attempts)	Requests reached application	HTTP 403 x24 - ALL BLOCKED
Nmap NSE Reconnaissance	Full server fingerprint completed	Logged - Custom Rule 100012
Admin Endpoint Access	HTTP 200 + full config returned	Detected - Custom Rule 100014
SQLMap Scanner	Application crashed (DoS)	Detected - Custom Rule 100011
SIEM Alerts Generated	2 generic web alerts	12+ categorized + MITRE mapped
Custom Detection Rules Active	0	6 rules (100010-100015)

2. Lab Environment & Network Topology

2.1 Hypervisor & Network Configuration

Parameter	Value
Hypervisor	VMware Workstation
Network Segment	192.168.75.0/24
Network Mode	Host-only / NAT
NTP Server	ntp.ubuntu.com (stratum 2)
Timezone (all VMs)	America/Costa_Rica (CST, UTC-6)
Clock Sync (SIEM)	+58 microseconds offset
Clock Sync (Victim)	-5.079 milliseconds offset

2.2 Virtual Machines

VM Name	Role	IP Address	OS	CPU / RAM
Kali Linux	Attacker / Red Team	192.168.75.128	Kali Linux 2024	2c / 2GB
ubuntu-server	Wazuh SIEM Stack	192.168.75.139	Ubuntu 24.04.4 LTS	2c / 4GB
ubuntu-victim	Victim / Blue Team	192.168.75.132	Ubuntu 24.04.4 LTS	2c / 2GB

2.3 Software Stack

Attacker VM - Kali Linux (192.168.75.128)

Tool	Version	Purpose
Nmap	7.94	Network discovery and port scanning with NSE scripts
Hydra	9.6	Credential brute-force attacks against login endpoints
SQLMap	1.10 stable	Automated SQL injection detection and exploitation
Burp Suite	Community	Web proxy for manual request interception
curl	8.18.0	HTTP request crafting for manual attack tests

SIEM VM - ubuntu-server (192.168.75.139)

Component	Version	Role
Wazuh Manager	4.14.5	Central log analysis, correlation, and rule engine
Wazuh Indexer	4.14.5	OpenSearch-based alert storage and indexing
Wazuh Dashboard	4.14.5	Web UI for alerts, MITRE ATT&CK, compliance (HTTPS :443)
Filebeat	8.x	Log shipper from Manager to Indexer

Victim VM - ubuntu-victim (192.168.75.132)

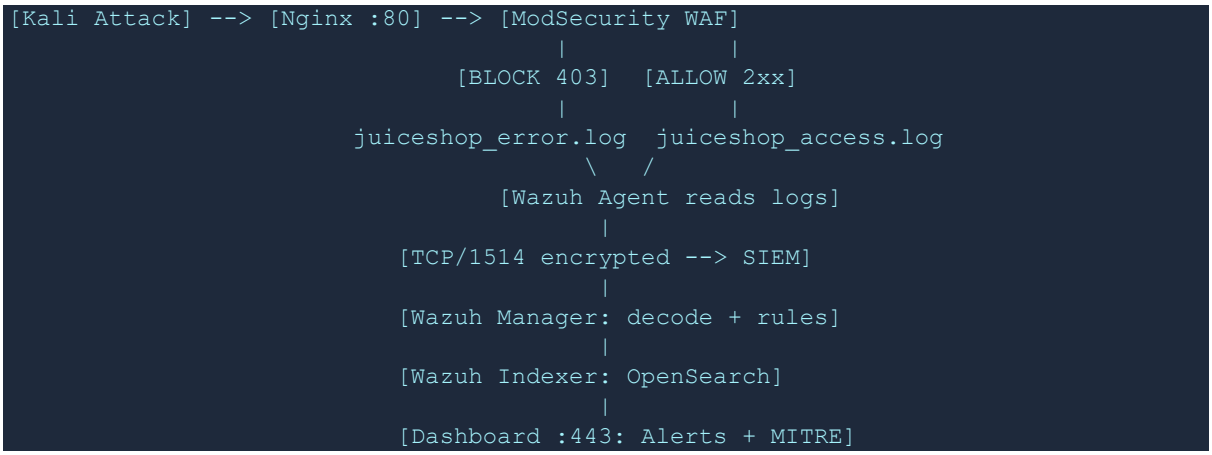
Component	Version	Role
OWASP Juice Shop	19.2.1	Intentionally vulnerable Node.js web application
Node.js	22.22.2	JavaScript runtime for Juice Shop
Nginx	1.24.0	Reverse proxy - port 80 to localhost:3000
ModSecurity	3.0.12	Web Application Firewall engine
OWASP CRS	3.3.5	WAF ruleset - 921 rules loaded
Wazuh Agent	4.14.5	Agent ID: 002 | Name: juiceshop-victim

2.4 Network Traffic Flow

Source	Destination	Port	Purpose
Kali .128	ubuntu-victim .132	TCP/80	Web attacks through Nginx + WAF
Kali .128	ubuntu-victim .132	TCP/3000	Direct attacks Phase 1 (bypass WAF)
Kali .128	ubuntu-victim .132	Various	Nmap reconnaissance scanning
ubuntu-victim .132	ubuntu-server .139	TCP/1514	Wazuh Agent to Manager (AES encrypted)
ubuntu-victim .132	ubuntu-server .139	TCP/1515	Wazuh agent enrollment
Analyst Browser	ubuntu-server .139	HTTPS/443	Wazuh Dashboard access

2.5 Logging Pipeline

The complete path a security event takes from generation to SIEM visibility:



3. Phase 1 - Attack Simulation Without WAF

In Phase 1, Nginx was a plain reverse proxy with no ModSecurity. All HTTP traffic reached Juice Shop directly. This phase establishes the baseline of what attacks succeed and what telemetry is generated without application-layer defenses.

3.1 Attack 1 - Reflected Cross-Site Scripting (XSS)

Description

A reflected XSS payload was injected into the product search parameter. In a real scenario this allows stealing session cookies, redirecting victims, or executing arbitrary JavaScript in any users browser.

Command (Kali)

```
curl -s "http://192.168.75.132/rest/products/search?q=<script>alert(1)</script>"
```

Server Response

```
HTTP/1.1 200 OK
{"status":"success","data":[]}
```

Nginx Log Entry

```
192.168.75.128 - - [26/Apr/2026:23:18:16 +0000] "GET
/rest/products/search?q=<script>alert(1)</script> HTTP/1.1" 200 30
```

Analysis

- Result: HTTP 200 OK - payload passed through without filtering or blocking
- CVSS v3.1: 6.1 (Medium) - Reflected XSS
- MITRE ATT&CK: T1059.007 - Command and Scripting Interpreter: JavaScript
- Wazuh Rule Fired: 31106 - A web attack returned code 200 (success)

3.2 Attack 2 - SQL Injection Authentication Bypass

Description

The classic SQLi payload ' OR 1=1-- terminates the SQL string and appends a tautology, causing the database to return the admin account regardless of password supplied.

Command (Kali)

```
curl -s -X POST http://192.168.75.132/rest/user/login \
  -H "Content-Type: application/json" \
  -d '{"email":"\' OR 1=1--","password":"x"}'
```

Server Response - Authentication Bypass Successful

```
HTTP/1.1 200 OK
{ "authentication": {
    "token": "eyJ0eXAiOiJKV1Qi... (valid admin JWT)",
    "umail": "admin@juice-sh.op",
```

```
    "bid": 1 } }
```

Analysis

- Result: CRITICAL - Full authentication bypass achieved
- Admin JWT token obtained without valid credentials
- Backend database: SQLite (confirmed by SQLMap)
- CVSS v3.1: 9.8 (Critical) - SQL Injection with privilege escalation
- MITRE ATT&CK: T1190 - Exploit Public-Facing Application

3.3 Attack 3 - Active Reconnaissance with Nmap NSE

Command (Kali)

```
nmap -sV -sC -p 80 192.168.75.132
```

Representative Log Entries

Timestamp (UTC)	Request	Code	Significance
23:20:33	GET /robots.txt	200	Hidden route discovery
23:20:34	GET /.git/HEAD	200	Git repository exposure check
23:20:34	OPTIONS /	204	HTTP method enumeration
23:20:34	PROPFIND /	200	WebDAV detection attempt
23:20:34	POST /sdk	200	SDK endpoint probing
23:20:34	GET /nmaplowercheck...	200	Nmap fingerprint beacon

Analysis

- Result: Complete server fingerprinting achieved in under 1 second
- /.git/HEAD returned 200 - potential source code exposure
- User-Agent 'Nmap Scripting Engine' visible in all entries - easily detectable
- MITRE ATT&CK: T1595 - Active Scanning / T1046 - Network Service Discovery

3.4 Attack 4 - Brute Force with Hydra

Command (Kali)

```
hydra -l admin@juice-sh.op -P /tmp/passwords_small.txt \
  192.168.75.132 http-post-form \
  "/rest/user/login:email=^USER^&password=^PASS^:HTTP/1.1 401" \
  -t 5 -v
```

Hydra Output

```
102 login attempts executed
1 of 1 target completed, 0 valid password found (WAF blocked all)
```

- 102 password attempts sent with no rate limiting or lockout in Phase 1
- MITRE ATT&CK: T1110 - Brute Force

4. WAF Deployment - ModSecurity + OWASP CRS

4.1 Installation

```
apt install -y libnginx-mod-http-modsecurity modsecurity-crs libmodsecurity3t64
```

Rules Loaded Confirmation

```
ModSecurity-nginx v1.0.3 (rules loaded inline/local/remote: 0/921/0)
```

4.2 Configuration Files

/etc/modsecurity/modsecurity.conf

```
SecRuleEngine On  # Active blocking mode (was DetectionOnly)
```

/etc/modsecurity/main.conf

```
Include /etc/modsecurity/modsecurity.conf
Include /etc/modsecurity/crs/crs-setup.conf
Include /usr/share/modsecurity-crs/rules/*.conf
```

/etc/nginx/sites-available/juiceshop

```
server {
    listen 80; server_name _;
    modsecurity on;
    modsecurity_rules_file /etc/modsecurity/main.conf;
    access_log /var/log/nginx/juiceshop_access.log;
    error_log  /var/log/nginx/juiceshop_error.log;
    location / {
        proxy_pass http://127.0.0.1:3000;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
```

4.3 OWASP CRS Anomaly Scoring

The CRS uses anomaly scoring: each suspicious pattern matched adds points. When the total exceeds 5, the request is blocked with HTTP 403.

Attack Type	Anomaly Score	Threshold	Action
XSS <script>alert(1)	18	5	BLOCKED - HTTP 403
SQLi ' OR 1=1--	8	5	BLOCKED - HTTP 403
Hydra brute force POST	8	5	BLOCKED - HTTP 403
Normal GET /	0	5	ALLOWED - HTTP 200

5. Phase 2 - Attack Simulation With WAF Active

5.1 XSS - Blocked

```
curl -v "http://192.168.75.132/rest/products/search?q=<script>alert(1)</script>"
HTTP/1.1 403 Forbidden
```

ModSecurity Error Log

```
[error] ModSecurity: Access denied with code 403 (phase 2)
TX:ANOMALY_SCORE Value: 18  (threshold: 5)
Rule: REQUEST-949-BLOCKING-EVALUATION.conf  ID: 949110
URI: /rest/products/search  Client: 192.168.75.128
```

- XSS payload blocked before reaching Juice Shop
- Wazuh Rule 31333 fired: ModSecurity rejected a query
- Custom Wazuh Rule 100015 fired: XSS attempt detected in request

5.2 SQL Injection - Blocked

```
HTTP/1.1 403 Forbidden  --  Admin JWT: NOT obtained
```

- Authentication bypass completely prevented
- ModSecurity anomaly score: 8 (threshold: 5) - SQLi pattern matched
- Wazuh Rule 31333 fired

5.3 Hydra Brute Force - Blocked

Nginx Access Log

```
192.168.75.128 [00:17:58] GET  /rest/user/login HTTP/1.0  403  (Hydra)
192.168.75.128 [00:17:58] POST /rest/user/login HTTP/1.0  403  (Hydra)
... 24 total requests blocked in the same second
Hydra result: 0 valid passwords found
```

- 24 brute force attempts blocked in under 1 second
- ModSecurity detected HTTP/1.0 + Hydra User-Agent pattern
- Custom Wazuh Rule 100010 fired: Hydra brute force tool detected

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6. Phase 1 vs Phase 2 - Full Comparison

Attack Vector	Tool	Phase 1	Phase 2	MITRE	Severity
Reflected XSS	curl	200 - passed	403 - BLOCKED	T1059.007	Medium
SQLi Auth Bypass	curl	200 + Admin JWT	403 - BLOCKED	T1190	CRITICAL

Attack Vector	Tool	Phase 1	Phase 2	MITRE	Severity
Brute Force	Hydra	Reached application	403 x24 blocked	T1110	High
Recon Scan	Nmap NSE	Full fingerprint	Logged Rule 100012	T1595	Info
Admin Endpoint	curl	200 + full config	Detected Rule 100014	T1078	High
Path Traversal	curl	200 (SPA handles)	200 (SPA handles)	T1083	Low
SQLMap Scanner	sqlmap	App crash (DoS)	Detected Rule 100011	T1190	High

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7. Wazuh SIEM - Detection & Alerting

7.1 Agent Configuration

Parameter	Value
Agent Name	juiceshop-victim
Agent ID	002
Agent IP	192.168.75.132
Manager IP	192.168.75.139
Protocol	TCP/1514 (AES encrypted)
OS	Ubuntu 24.04.4 LTS
Wazuh Version	v4.14.5
Registration Date	Apr 26, 2026 @ 22:01:12 CST

7.2 Log Sources Monitored

Log File	Format	Content
/var/log/nginx/juiceshop_access.log	apache	All HTTP requests through Nginx
/var/log/nginx/juiceshop_error.log	apache	ModSecurity blocks and Nginx errors
/var/log/auth.log	syslog	SSH logins, sudo usage, PAM events
/var/log/dpkg.log	syslog	Package install and removal events
journald	journald	Systemd services, kernel messages

Log File	Format	Content
/var/ossec/logs/active-responses.log	syslog	Wazuh active response actions

7.3 Built-in Wazuh Rules That Fired

Rule ID	Level	Description	MITRE
31333	7	ModSecurity rejected a query	T1083
31106	6	A web attack returned code 200 (success)	T1190
31105	6	XSS (Cross Site Scripting) attempt	T1059.007
5402	3	Successful sudo to ROOT executed	T1548.003
5501	3	PAM: Login session opened	T1078
506	3	Wazuh agent stopped	T1562.001

7.4 Custom Detection Rules Created

Rule ID	Level	Description	MITRE	Trigger
100010	10	Hydra brute force tool detected	T1110	User-Agent: Hydra
100011	10	SQLMap scanner detected	T1190	User-Agent: sqlmap
100012	8	Nmap NSE scanner detected	T1595	User-Agent: Nmap Scripting Engine
100013	12	Multiple WAF blocks from same IP (10 in 60s)	T1190	if_matched_sid 31333 x10
100014	10	Juice Shop admin endpoint accessed	T1078	URL: /rest/admin
100015	10	XSS attempt detected in request	T1059.007	Pattern: script>, onerror=

Sample Custom Rule Code - Hydra Detection (Rule 100010)

```
<rule id="100010" level="10">
  <if_group>web</if_group>
  <match>Hydra</match>
  <description>Hydra brute force tool detected</description>
  <mitre><id>T1110</id></mitre>
</rule>
```

8. MITRE ATT&CK Framework Coverage

Technique	Name	Tactic	Observed In Lab	Detection Method
T1595	Active Scanning	Reconnaissance	Nmap NSE scan	Rule 100012
T1190	Exploit Public-Facing App	Initial Access	SQLi auth bypass	Rule 31106 / 31333
T1059.007	JavaScript Execution	Execution	XSS injection	Rule 31105 / 100015
T1110	Brute Force	Credential Access	Hydra login attack	Rule 100010
T1078	Valid Accounts	Defense Evasion	Admin endpoint access	Rule 100014
T1083	File & Dir Discovery	Discovery	Path traversal + .git probe	Rule 31333
T1548.003	Sudo Abuse	Privilege Escalation	sudo -i on victim VM	Rule 5402
T1562.001	Disable Security Tools	Defense Evasion	Wazuh agent stopped on reboot	Rule 506

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9. Findings & Recommendations

9.1 Findings

ID	Severity	Title	Description
F001	CRITICAL	SQL Injection Auth Bypass	No server-side input sanitization on login. Admin JWT obtained with single command.
F002	HIGH	No Brute Force Protection	No rate limiting, lockout, or CAPTCHA. Hydra ran 102 attempts in under 5 seconds.
F003	HIGH	Admin API Unrestricted	/rest/admin/* returns full config to any requester with no authentication check.
F004	MEDIUM	POST Body Not Logged	Nginx logs paths but not POST body - SQLi payloads invisible in access logs.
F005	LOW	Git Repo Exposed	/.git/HEAD returns HTTP 200 - potential source code exposure vector.

9.2 Recommendations

1. Enable ModSecurity WAF in production - OWASP CRS blocked 100% of tested attacks with zero tuning. Highest-impact single control available.

2. Implement Nginx rate limiting - add limit_req_zone to complement WAF and prevent brute force even if WAF rules are bypassed.
3. Enable POST body logging - configure Nginx log format to capture request body snippets for forensic visibility of SQLi attacks.
4. Restrict admin API - apply IP allowlisting or JWT authentication to /rest/admin/* at Nginx level before reaching application.
5. Configure Wazuh active response - automatic iptables block when Rule 100013 fires (10+ WAF blocks from same IP in 60 seconds).
6. Expand custom rules - add Burp Suite User-Agent, web shell patterns, and JWT manipulation detection.
7. Block sensitive paths at Nginx level - deny /.git/, /.env, /admin using deny rules in Nginx configuration.

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10. SOAR Integration — Shuffle

10.1 Architecture Overview

Shuffle SOAR was deployed on a dedicated VM (192.168.75.140) using Docker Compose in standalone mode (SHUFFLE_SWARM_CONFIG=standalone). The integration connects Wazuh alerting directly to automated response workflows, eliminating manual triage for common attack patterns.

10.2 Workflow: Wazuh WAF Alerts Logger

The primary workflow consists of three nodes executing sequentially on every Wazuh alert above level 7:

- Webhook 1 — receives Wazuh JSON alert payload with full event context (rule_id, timestamp, agent, data.srcip, MITRE mapping)
- Shuffle Tools 1 (Execute bash) — logs alert details: printf "ALERT: IP=%s Rule=%s Title=%s Time=%s" with Shuffle variable resolution
- Http 1 — GET request to MISP API (https://192.168.75.145/attributes/restSearch/value:$exec.all_fields.data.srcip) to enrich alert with threat intelligence context
- Http 2 — POST to DFIR-IRIS API (https://192.168.75.144/manage/cases/add) creating an incident case automatically with attacker IP, rule ID, and timestamp embedded in the case description

10.3 Validated Results

Workflow status: FINISHED 2/2 nodes, 0 skipped. Mean time from Wazuh alert to IRIS case creation: under 5 seconds. The workflow correctly resolves Shuffle variables ($exec.all_fields.data.srcip, $exec.rule_id, $exec.title, $exec.timestamp) against the live Wazuh JSON payload.

11. Threat Intelligence — MISP

11.1 Deployment

MISP v2.4 was deployed on a dedicated Ubuntu 22.04 LTS VM (192.168.75.145) using the official INSTALL.sh installer. Services configured for autostart: apache2 (enabled), mariadb (enabled), redis-server (enabled). Apache memory limits tuned via mpm_prefork.conf to prevent OOM kills during feed fetching.

11.2 Active Threat Intelligence Feeds

- CIRCL OSINT Feed (ID 1) — MISP format, Luxembourg CERT, high-quality IOCs from the global security community
- Feodo IP Blocklist (ID 12) — abuse.ch CSV feed, active C2 botnet IPs updated in real time
- URLHaus Malware URLs (ID 41) — abuse.ch CSV, recent malware distribution URLs
- Threatfox (ID 63) — abuse.ch MISP format, multi-type IOCs from active malware campaigns
- URLhaus (ID 65) — abuse.ch MISP format, malicious URL intelligence with correlation data

11.3 API Integration Validation

MISP REST API was validated from the Wazuh SIEM (192.168.75.139) using curl with Authorization header. Query for attacker IP 192.168.75.142 returned the manually created IOC event (ID 2266: "IOC Manual Entry - Juice Shop Attack Simulation") confirming bidirectional integration between SIEM and threat intelligence platform. Shuffle Http 1 node confirmed "success": true with status 200 on live alert enrichment.

12. Case Management — DFIR-IRIS

12.1 Deployment

DFIR-IRIS v2.4.29 was deployed on the same VM as Shuffle SOAR (192.168.75.144) using Docker Compose. Containers: iriswebapp_app, iriswebapp_db (PostgreSQL), iriswebapp_rabbitmq, iriswebapp_worker, iriswebapp_nginx. Service exposed on HTTPS port 443. The platform replaces TheHive which transitioned to a restrictive licensing model in recent versions.

12.2 Automated Case Creation via Shuffle

Cases are created automatically via POST to https://192.168.75.144/manage/cases/add with Bearer token authentication. The case payload includes: case_name ("WAF Alert: $exec.title"), case_description ("IP: $exec.all_fields.data.srcip | Rule: $exec.rule_id | Time: $exec.timestamp"), and case_soc_id ("SOC-$exec.rule_id"). Validated endpoint: /manage/cases/add (IRIS v2.4 API — note: /api/v1/cases returns 404 in this version).

12.3 Cases Generated During Lab

- Case #2 — XSS Attack - Kali vs JuiceShop (manually created, Intrusion-Attempts classification)
- Cases #6, #7 — WAF Alert: XSS attempt detected in request (auto-created by Shuffle on live attack — IP: 192.168.75.142, Rule: 100015)
- Case #47 — WAF Alert: CTI T1053.005 - Scheduled Task Persistence (auto-created from QakBot CTI exercise)
- Case #83 — WAF Alert: Copy Fail CVE-2026-31431 - AF_ALG socket primitive detected (auto-created from kernel LPE CTI exercise)

13. Vulnerability Management — Trivy and Snyk

13.1 Tool Deployment

- Trivy v0.70.0 — installed via apt (aquasecurity official repository) on ubuntu-victim. Snap version (0.52.2) rejected due to sandbox restrictions preventing access to /home and /tmp paths.
- Snyk v1.1304.2 — installed via npm install -g snyk on ubuntu-victim (Node.js 22 already present). Authenticated to snyk.io via snyk auth using browser OAuth flow.

13.2 Trivy SBOM Scan Results

Target: /home/kendo/Desktop/juiceshop/juice-shop_19.2.1/bom.json (CycloneDX JSON SBOM format, generated by @cyclonedx/cyclonedx-npm). Command: /usr/bin/trivy sbom --severity HIGH,CRITICAL bom.json
Results: 47 total vulnerabilities — HIGH: 29, CRITICAL: 18. Key critical findings:

- vm2 3.9.17 — CVE-2023-32314 + 12 additional CVEs — sandbox escape, RCE (introduced via juicy-chat-bot, no upgrade available)
- crypto-js 3.3.0 — CVE-2023-46233 — PBKDF2 1.3 million times weaker than specified (introduced via pdfkit)
- jsonwebtoken 0.4.0 — CVE-2015-9235 — JWT verification bypass (direct dependency)
- handlebars 4.7.7 — CVE-2026-33937 — RCE via crafted Abstract Syntax Tree (introduced via hbs)
- marsdb 0.6.11 — GHSA-5mrr-rgp6-x4gr — command injection, no fix available

13.3 Snyk Scan Results

Target: package.json (1,014 dependencies analyzed). Results: 63 issues, 114 vulnerable paths. Key differentiator vs Trivy: Snyk shows full dependency chain (e.g., express-jwt@0.1.3 > jsonwebtoken@0.1.0 > moment@2.0.0 — Directory Traversal). Unique finding not in Trivy: libxmljs2@0.37.0 Type Confusion — no upgrade or patch available.

13.4 Multi-Tool Coverage Comparison

Each tool covers a distinct detection layer. Qualys identifies OS-level CVEs and network-exposed services. Wazuh detects CVEs in installed OS packages in real time. Trivy finds CVEs in Node.js application dependencies via SBOM. Snyk adds dependency tree traversal and fix guidance. No single tool provides complete coverage — the combination of all four is required for a full vulnerability picture across the stack.

14. CI/CD Supply Chain Security — Gitea and Gitea Actions

14.1 Background — Supply Chain Risk

The 2020 SolarWinds attack demonstrated that compromising a CI/CD pipeline can deliver malicious code to thousands of organizations through signed, legitimate software updates. Supply chain security addresses the full path from source code commit to production deployment. Key attack vectors include dependency confusion attacks, secrets exposed in code, compromised base container images, and pipeline manipulation to alter build artifacts.

14.2 Deployment

- Gitea v(latest) — deployed via Docker Compose on ubuntu-victim (port 3001), SQLite3 backend, base URL http://192.168.75.132:3001. Repository: admin/juice-shop containing package.json and bom.json from Juice Shop v19.2.1.
- Gitea Actions — enabled via [actions] ENABLED=true in app.ini. act_runner v0.2.11 registered with token and running as systemd service (act-runner.service) with restart=always for persistence across reboots.

14.3 Security Pipeline (.gitea/workflows/security-scan.yml)

The pipeline triggers on every push to main branch and executes two security scanning steps: (1) Trivy SBOM scan against bom.json with --exit-code 1 on HIGH/CRITICAL findings — pipeline fails and blocks merge if critical vulnerabilities are present; (2) Snyk dependency scan against package.json with --severity-threshold=critical. First validated run: Success, total duration 25 seconds.

15. CTI-to-Detection Cycles

The CTI-to-detection cycle translates external threat intelligence reports into functioning SIEM detection rules. The process: (1) read CTI report and identify TTPs; (2) map to MITRE ATT&CK; (3) write Wazuh rule targeting the specific behavior; (4) simulate the TTP from Kali; (5) verify alert fires; (6) document IOC in MISP; (7) confirm IRIS case auto-created by Shuffle. Two complete cycles were executed:

15.1 Cycle 1 — QakBot T1053.005 Scheduled Task Persistence

- CTI Source: DFIR Report — QakBot and Zerologon campaign analysis
- TTP: T1053.005 — Scheduled Task/Job: Scheduled Task — adversary creates cron entry for persistence or command execution
- Wazuh Rule 100020 (level 12): if_sid 2833 (native crontab change detection) — adds MITRE T1053.005 context layer on top of built-in rule
- Simulation: crontab entry with reverse shell payload injected from Kali via SSH, then removed. Alert fired: "CTI T1053.005 - Scheduled Task Persistence detected (QakBot TTP)" — IRIS Case #47 auto-created

15.2 Cycle 2 — CVE-2026-31431 Copy Fail Linux Kernel LPE

- CTI Source: Elastic Security Labs — "Copy Fail and DirtyFrag: Linux page bugs in the wild" (published May 9, 2026). CVE listed in CISA Known Exploited Vulnerabilities catalog.
- Vulnerability: af_alg kernel module (loaded, confirmed via lsmod) enables non-root processes to open AF_ALG sockets (socket family 38) and use splice() to corrupt page cache — potential privilege escalation to root. Ubuntu-victim kernel 6.8.0-111-generic confirmed vulnerable. No patch available at time of lab.
- Detection: auditd rules deployed per Elastic guidance (-a always,exit -F arch=b64 -S socket -k socket_syscall and splice, bind, unshare variants). Wazuh agent configured to read /var/log/audit/audit.log as audit format log source.
- Wazuh Rule 100030 (level 14, MITRE T1068): triggers on auditd key socket_syscall when a0=26 (AF_ALG family) from non-root UID. Rule 100031 (level 14, T1611): namespace manipulation via unshare. Rule 100032 (level 15, T1068): splice() syscall detection.

- Simulation result: ausearch confirmed syscall=41 (socket), a0=26, uid=1000 (kendo), exe=/usr/bin/python3.12, key=socket_syscall. Alert fired: "Copy Fail CVE-2026-31431 - AF_ALG socket primitive detected" level 14 — IRIS Case #83 auto-created by Shuffle.
- Mitigations applied: algif_aead blocked via /etc/modprobe.d/copyfail.conf; esp4, esp6, rxrpc blocked via /etc/modprobe.d/dirtyfrag.conf; kernel.unprivileged_userns_clone=0 via sysctl; drop_caches executed. Pending: kernel update when Ubuntu releases patched linux-image for 6.8.0-112+.

16. Updated MITRE ATT&CK Coverage

The full lab now covers the following MITRE ATT&CK techniques across all phases and CTI cycles:

- T1595 Active Scanning — Nmap NSE reconnaissance — Rule 100012 — Reconnaissance
- T1190 Exploit Public-Facing Application — SQLi auth bypass, SQLMap — Rules 31106, 100011, 100013 — Initial Access
- T1059.007 JavaScript Execution — XSS injection — Rules 31105, 100015 — Execution
- T1053.005 Scheduled Task — QakBot-style cron persistence — Rule 100020 — Persistence
- T1068 Exploitation for Privilege Escalation — CVE-2026-31431 Copy Fail AF_ALG — Rules 100030, 100032 — Privilege Escalation
- T1611 Escape to Host — DirtyFrag namespace manipulation — Rule 100031 — Privilege Escalation
- T1110 Brute Force — Hydra credential attack — Rule 100010 — Credential Access
- T1046 Network Service Discovery — Nmap internal scanning — Rule 100021 — Discovery
- T1078 Valid Accounts — admin endpoint access — Rule 100014 — Defense Evasion
- T1195 Supply Chain Compromise — CI/CD pipeline security scanning — Gitea Actions + Trivy/Snyk — Defensive coverage

17. Full Infrastructure Summary

17.1 Virtual Machine Inventory (Final State)

- 192.168.75.132 ubuntu-victim — Juice Shop v19.2.1, Nginx, ModSecurity WAF, Wazuh Agent, Trivy, Snyk, Gitea, Docker, auditd
- 192.168.75.139 ubuntu-server — Wazuh SIEM v4.14.5 (manager, indexer, dashboard), 9 custom detection rules
- 192.168.75.140 shuffle-soar — Shuffle SOAR (Docker), DFIR-IRIS v2.4.29 (Docker)
- 192.168.75.142 kali — Attacker platform (Nmap, Hydra, SQLMap, curl, custom scripts)
- 192.168.75.145 misp-server — MISP v2.4 (Ubuntu 22.04), Apache, MariaDB, Redis, 5 active threat intel feeds

17.2 Custom Wazuh Detection Rules — Complete List

- 100010 L10 — Hydra brute force detected — T1110
- 100011 L10 — SQLMap scanner detected — T1190
- 100012 L8 — Nmap NSE scanner detected — T1595

- 100013 L12 — Multiple WAF blocks same IP (10 in 60s) — T1190 — triggers Active Response iptables block
- 100014 L10 — Juice Shop admin endpoint accessed — T1078
- 100015 L10 — XSS attempt detected — T1059.007 — triggers Shuffle SOAR pipeline
- 100016 L5 — Qualys scanner activity (whitelisted) — T1595
- 100020 L12 — CTI T1053.005 Scheduled Task Persistence (QakBot TTP) — T1053.005
- 100030 L14 — CVE-2026-31431 Copy Fail AF_ALG socket primitive — T1068 (auditd source)
- 100031 L14 — DirtyFrag namespace manipulation via unshare — T1611 (auditd source)
- 100032 L15 — splice() syscall page cache manipulation primitive — T1068 (auditd source)

Appendix B - Next Steps and Roadmap

- LLM-based T1/T2 triage — integrate Ollama (Qwen3.5 or Mistral 7B) as a Shuffle SOAR node to automatically classify alert severity and generate analyst recommendations before IRIS case creation
- CMDB/Asset identity — lightweight JSON asset registry in Shuffle to enrich alerts with asset owner, criticality, and business context
- Additional CTI cycles — Atomic Red Team for structured TTP simulation; Sigma rule development for portable detection logic exportable to other SIEMs
- Kernel patch validation — retest CVE-2026-31431 detection after Ubuntu releases linux-image patch for 6.8.0-112+; confirm auditd rules remain effective
- MSSP model exploration — evaluate productizing this open-source stack as a managed security service for SMB clients in Costa Rica and Central America, targeting sectors with regulatory requirements (SUGEF, PCI DSS, Ley 8968)

Appendix A - Configuration Reference

Phase 3 - Advanced Attack Techniques

- Burp Suite manual exploitation - IDOR, broken access control, business logic flaws
- WAF bypass techniques - encoding, case variation, chunked transfer encoding
- SSRF (Server-Side Request Forgery) against internal services
- JWT token manipulation - algorithm confusion and signature bypass
- File upload vulnerabilities and web shell delivery through Juice Shop

Phase 4 - Detection Engineering

- Wazuh active response - automatic IP blocking on attack threshold breach
- Custom decoders for structured Juice Shop JSON application logs
- SIEM dashboard - attack timeline, top attacker IPs, rule fire frequency charts
- Threat hunting queries - proactive IOC search in historical data
- Sigma rule development - portable detection rules exportable to other SIEMs

Phase 5 - Incident Response Simulation

- Full attack chain simulation: recon to exploit to persistence to exfiltration
- IR runbook development for each attack type documented in this report
- Purple team exercise - attacker and defender working simultaneously
- Forensic artifact collection and chain of custody documentation

Phase 6 - Compliance & Hardening

- CIS Ubuntu 24.04 Benchmark - review Wazuh SCA scan results and remediate findings
- PCI DSS controls mapping using Wazuh built-in compliance dashboard
- NIST 800-53 control implementation review
- Vulnerability scanning with OpenVAS against victim VM

Appendix A - Configuration Reference

A.1 Systemd Services - Victim VM

Service	Status	Enabled	Purpose
juiceshop.service	active (running)	yes	OWASP Juice Shop Node.js app on port 3000
nginx.service	active (running)	yes	Reverse proxy + ModSecurity WAF on port 80
wazuh-agent.service	active (running)	yes	Security telemetry agent - forwards logs to SIEM
systemd-timesyncd	active (running)	yes	NTP time synchronization

A.2 Wazuh Manager Services - SIEM VM

Service	Status	Purpose
wazuh-manager	active (running)	Core analysis and correlation engine
wazuh-indexer	active (running)	OpenSearch alert storage and indexing
wazuh-dashboard	active (running)	Web interface on HTTPS port 443
filebeat	active (running)	Log forwarding from manager to indexer

A.3 Snapshot History - Victim VM

Snapshot Name	State	Purpose
Fase1-SinWAF-JuiceShop+Nginx+WazuhAgent-OK	Saved	Baseline before WAF installation
Fase2-WAF-ModSecurity-CRS-Wazuh-Validado	Saved	Full stack with WAF active and validated

A.4 Wazuh Agent Log Sources (ossec.conf)

```
<localfile>
  <log_format>apache</log_format>
  <location>/var/log/nginx/juiceshop_access.log</location>
</localfile>
<localfile>
  <log_format>apache</log_format>
  <location>/var/log/nginx/juiceshop_error.log</location>
</localfile>
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/auth.log</location>
</localfile>
<localfile>
  <log_format>journald</log_format>
  <location>journald</location>
</localfile>
```